

DTS: Discriminant Truthfulness Scoring for Gradient-Free Hallucination Detection

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Abstract

Large language models can hallucinate with the same fluent confidence they bring to correct answers, making reliable hallucination detection essential for trustworthy deployment. Existing detectors based on LLM hidden states often rely on cosine or linear readouts that ignore the anisotropic structure of the representation space. We introduce DTS (Discriminant Truthfulness Scoring), a lightweight, gradient-free detector that whitens hidden activations using within-class covariance and scores responses with a closed-form Fisher discriminant. Across Qwen-2.5-7B and Llama-3.1-8B on four QA datasets, DTS ranks first in five of eight settings and second in the remaining three. Whitening improves over cosine scoring by 7.8 AUROC points on average, identifying the readout as a major bottleneck. When truthful and hallucinated representations overlap, class-mean steering provides additional gains without learning a steering vector. DTS requires only a single forward pass and closed-form computation, making it substantially cheaper than sampling- and gradient-based alternatives, while also extending effectively to long-form generation.

1 Introduction

Large language models now draft code, answer technical questions, and inform decisions in settings where a wrong answer carries real cost. They also hallucinate, stating false claims as confidently as true ones, which makes such errors hard to catch from the text alone. A detector that flags these failures is a prerequisite for trusting the output. We restrict attention to detectors that use only the generating model, with no retrieval, external judge, or second model (Huang et al., 2023), since these run wherever the model already runs. Prior work clusters into four families, each strong somewhere and fragile elsewhere.

Output-confidence methods read perplexity, predictive entropy, or a model’s stated confidence (Kadavath et al., 2022). One forward pass buys them, but they report high confidence exactly when the model hallucinates fluently. *Sampling* methods compare several generations instead, via self-consistency (Manakul et al., 2023) or semantic entropy (Kuhn et al., 2023; Farquhar et al., 2024); the extra samples buy accuracy, yet a model that is confidently and repeatedly wrong defeats them. The last two families work in representation space. *Internal-state* methods read hidden activations directly, through supervised probes (Azaria and Mitchell, 2023) or unsupervised geometry (Chen et al., 2024; Du et al., 2024; Burns et al., 2023). Most need only one forward pass, but they compare truthful and hallucinated states in the model’s native geometry, which is strongly anisotropic; a high-variance direction carrying no truthfulness signal can then swamp the comparison. *Steering* methods reshape the activations so the classes pull apart (Li et al., 2023; Rimskey et al., 2024). TSV (Park et al., 2025) learns such an intervention from labeled data by gradient descent, then scores the reshaped states by raw similarity, leaving the gain tangled up with the training spent to obtain the direction.

These methods share two assumptions, and we follow each to its consequence. The first concerns the readout. Cosine similarity treats every direction as equally informative, but hidden states are not isotropic: a real gap between class means can lie along a high-variance nuisance axis and vanish in the score. The standard fix is to whiten the space by the within-class covariance before measuring distance, which is the Mahalanobis distance of an answer to the two class means (Fisher, 1936; Lee et al., 2018). That discriminant has a closed form in the labeled activations, with no gradient descent and no learned steering vector; the only choices are a layer and a shrinkage coefficient, both set

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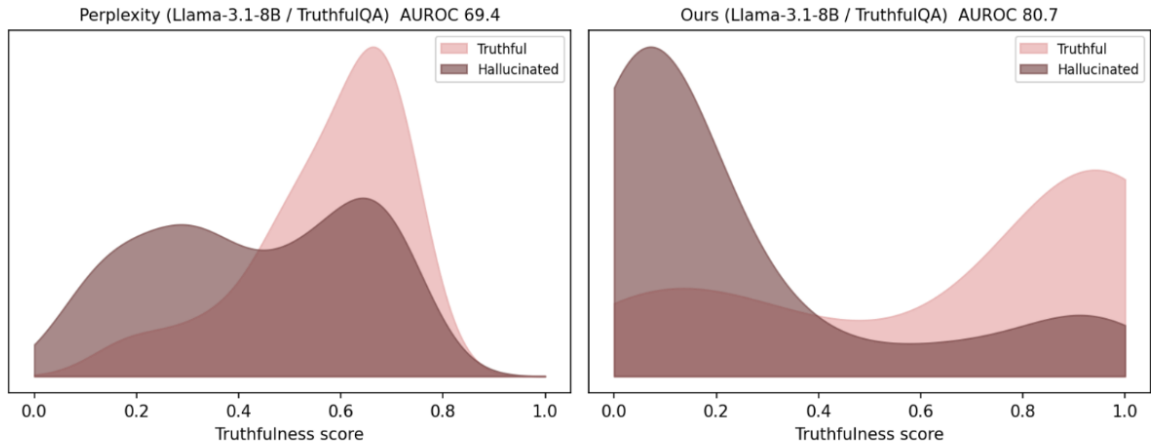


Figure 1: Truthfulness-score distributions on Llama-3.1-8B / TruthfulQA. A perplexity baseline (left) leaves truthful and hallucinated answers overlapping (AUROC 69.4); our closed-form discriminant (right) separates them (AUROC 80.7). Scores are Platt-calibrated to $[0, 1]$. The two panels come from a separate visualisation pipeline, so their AUROCs differ by up to two points from the held-out test figures in Table 1.

by cross-validation, and a one-dimensional logistic calibration onto a probability. We call the detector DTS (Discriminant Truthfulness Scoring). It uses class labels, like the supervised methods it competes with, but estimates the direction in closed form and fits nothing beyond the scalar calibrator. The second assumption ties reshaping the representation to the training used to find the reshaping. When the steering vector is learned by gradient descent, a gain could come from moving the activations or from the labels spent learning the move; we pull the two apart.

The headline result is that the readout, not the steering, is what limits these detectors. Where the activations are already separable, the closed-form discriminant matches the best reported detectors. On Qwen-2.5-7B it reaches 90.1 AUROC on TruthfulQA and 91.6 on TriviaQA, 1.4 and 7.4 points above the fully-supervised TSV † . Even on the harder Llama-3.1-8B, where a perplexity baseline cannot tell the two classes apart, the discriminant pulls them well apart (Figure 1). Its remaining shortfall there is geometric, not algorithmic: Qwen’s truthful and hallucinated states separate cleanly ($d'=1.97$) while Llama’s sit almost on top of each other ($d'=1.29$; Figure 2), and no linear readout can split clouds that overlap. Steering helps in this regime, and a direction computed directly from the class means performs comparably to a trained steering vector. In contrast, a random direction of the same norm provides no benefit, suggesting that the useful steering signal is already

encoded in the model’s activation geometry rather than learned by the detector.

We make the following contributions.

- We trace the limit of hidden-state detectors to the readout rather than the steering: a closed-form whitened discriminant beats a trained cosine readout and matches competitive supervised detectors on the same labels, training nothing of its own.
- We give a gradient-free steering vector that equals a trained one, and tie the need for steering to the representation geometry (d' , overlap), so “does steering help” becomes a property of the model.
- Across two models and four datasets, DTS matches or exceeds the fully-supervised TSV † on five of eight settings and ranks first or second on every one, with no gradient training.

2 Related Work

Hallucination from hidden states. These methods read internal activations rather than generated text. Supervised probes train a classifier on activations labelled truthful or hallucinated (Azaria and Mitchell, 2023; Liu et al., 2024). Unsupervised methods read geometry directly: INSIDE uses the covariance eigenspectrum of several sampled response embeddings (Chen et al., 2024), HaloScope fits an SVD subspace over unlabelled generations (Du et al., 2024), CCS finds a direction consistent

under negation (Burns et al., 2023), and others use attention-map spectra (Binkowski et al., 2025) or the unembedding matrix (Hu et al., 2025). Most read the generating model’s activations in a single forward pass, the property we keep, though INSIDE needs several samples to estimate its covariance. Each scores by a cosine, linear probe, or projection magnitude, and none use the within-class covariance of the two classes, which, as we show, carries most of the separating signal.

Discriminant analysis and steering. Our readout is textbook. Fisher’s discriminant separates two classes along $S_W^{-1}(\mu_+ - \mu_-)$, whitening the space so within-class variation is isotropic before measuring the distance between class means (Fisher, 1936); the same construction underlies a standard Mahalanobis out-of-distribution detector (Lee et al., 2018). Replacing the covariance with the identity recovers the plain mass-mean direction of linear truth probes (Marks and Tegmark, 2024; Li et al., 2023); we use the full whitened form. A separate line edits activations to change behaviour (Zou et al., 2023): ITI adds a fixed vector to attention heads (Li et al., 2023), CAA builds it as a mean difference over contrasting inputs (Rimsky et al., 2024), and TSV *trains* a steering vector for hallucination detection (Park et al., 2025). Our steering step uses the same contrastive direction in closed form, with no optimisation; we show it equals the trained vector and is needed only when the two distributions overlap.

3 Methodology

DTS (Discriminant Truthfulness Scoring) is a label-supervised, closed-form hallucination detector. The generator stays frozen throughout; detection reduces to finding the single linear direction in its activation space along which truthful and hallucinated answers separate best, then calibrating that separation into a probability. The model is never updated and the high-dimensional direction is closed form; a two-parameter calibrator is fit on the labelled development pool.

$$(q, a) \xrightarrow{\text{LLM}} \mathbf{h} \xrightarrow{\text{disc.}} s \xrightarrow{\text{calib.}} \text{Pr}(\text{truthful})$$

3.1 Setup

Let $\mathcal{M}$ be a frozen autoregressive model with L layers and hidden width D . For a pair (q_i, a_i) we form $x_i = [\text{PROMPT}(q_i); a_i]$ and read the hidden

state of the final answer token at every layer,

$$\mathbf{h}_i^{(\ell)} = [\mathcal{M}^{(\ell)}(x_i)]_{T_i} \in \mathbb{R}^D, \quad \ell = 1, \dots, L. \quad (1)$$

Under causal masking the last token T_i is the only position that attends to the full input, so it is a natural response-level representation. It is not assumed to be universally optimal: we compare it with mean and max pooling in §4.5. We keep all layers rather than fix one, because the depth at which truthfulness is linearly decodable shifts across models and tasks; the data chooses it (§3.3).

Truthfulness labels are derived by comparing each model answer a_i against a gold reference a_i^* using BLEURT (Sellam et al., 2020). Specifically,

$$y_i = \begin{cases} 1, & \text{BLEURT}(a_i, a_i^*) > \tau, \\ 0, & \text{otherwise.} \end{cases} \quad (2)$$

where τ is a fixed threshold. These labels are used only for fitting and evaluating the detector, and are never used at inference time. The labeled corpus is split once into a development pool $\mathcal{P}$ and a held-out test set $\mathcal{T}$; all modeling decisions are made exclusively on $\mathcal{P}$.

3.2 Discriminant Detector

We seek the projection of $\mathbf{h}^{(\ell)}$ that places the two classes farthest apart relative to their within-class spread, the Fisher criterion

$$\mathbf{w}^* = \arg \max_{\mathbf{w}} \frac{(\mathbf{w}^\top (m_1 - m_0))^2}{\mathbf{w}^\top S_w \mathbf{w}}, \quad (3)$$

with class means m_1, m_0 and pooled within-class covariance S_w . Its maximiser $\mathbf{w}^* \propto S_w^{-1}(m_1 - m_0)$ is closed form, so no iterative optimisation is needed.

PCA projection. Estimating S_w directly in $\mathbb{R}^D$ is ill-posed: with D in the thousands, often comparable to or larger than the per-class sample count, the pooled scatter is rank-deficient or badly conditioned. We therefore project the activations onto the top- p principal directions of the centred pool, giving coordinates $\mathbf{z}_i \in \mathbb{R}^p$ in which the covariance is full rank. During cross-validation, centering and PCA are fit on each fold’s training partition only.

Whitened discriminant. In this subspace we regularise the pooled scatter toward a scaled identity,

$$\hat{S}_w = S_w + \lambda \frac{\text{tr}(S_w)}{p} I_p. \quad (4)$$

Scaling the penalty by the mean eigenvalue keeps λ invariant to activation scale; its value is set by cross-validation (§3.3).

Whitening by $W = \hat{S}_w^{-1/2}$ makes within-class variation isotropic, after which the optimal separator is just the direction joining the class means. This yields a unit truthfulness axis and a scalar score,

$$\begin{aligned} \mathbf{d} &= \frac{W(m_1 - m_0)}{\|W(m_1 - m_0)\|}, \\ s_i &= \mathbf{d}^\top W \mathbf{z}_i \propto (m_1 - m_0)^\top \hat{S}_w^{-1} \mathbf{z}_i. \end{aligned} \quad (5)$$

The right-hand form is exactly the Fisher discriminant of Eq. (3), so whitening is a numerically stable route to the same solution. A logistic link fitted on the pool with class-balanced weights turns the standardized score $\tilde{s} = (s - \bar{s})/\text{sd}(s)$ into a probability,

$$\Pr(y=1 \mid \mathbf{h}^{(\ell)}) = \sigma(\beta_0 + \beta_1 \tilde{s}). \quad (6)$$

The pair (β_0, β_1) is the only thing fitted by optimisation; everything upstream is a closed-form statistic of the activations.

3.3 Layer and Hyperparameter Selection

The most decodable layer and appropriate discriminant subspace are unknown a priori, so we let the data choose them. We jointly select the layer ℓ , PCA dimension p , and shrinkage λ by stratified K -fold cross-validation on $\mathcal{P}$, maximising AUC,

$$(\ell^*, p^*, \lambda^*) = \arg \max_{\ell, p, \lambda} \frac{1}{K} \sum_{k=1}^K \text{AUC}_k(\ell, p, \lambda), \quad (7)$$

where AUC is threshold-free and stable under class imbalance. For every fold and candidate configuration, PCA, class means, covariance, whitening, score standardisation, and calibration are refit using only that fold’s training partition. After selection, these quantities are fit once on all of $\mathcal{P}$ using (ℓ^*, p^*, λ^*) , and the resulting detector is applied once to $\mathcal{T}$; uncertainty is quantified by a bootstrap standard error over B resamples. Further details and robustness analyses for these choices are provided in Appendix D.

3.4 Causal Steering

When truthful and hallucinated activations overlap, we reshape the representation by intervening on the residual stream at a single layer ℓ_s . The steering

direction is the class-mean difference in the original activation space,

$$\mathbf{v} = \mu_1^{(\ell_s)} - \mu_0^{(\ell_s)}, \quad (8)$$

where $\mu_c^{(\ell_s)} \in \mathbb{R}^D$ denotes the class- c mean activation. We add $\alpha \mathbf{v}$ to every token, $\tilde{\mathbf{h}}_t^{(\ell_s)} = \mathbf{h}_t^{(\ell_s)} + \alpha \mathbf{v}$, and refit the detector on layers $\{\ell_s + 1, \dots, L\}$. The steering layer ℓ_s and strength α are selected by cross-validated AUC on the pool (§4.4); the better of the steered and unsteered variants is then evaluated once on the held-out test set.

The complete DTS detection procedure, including model selection, refitting, held-out evaluation, and optional causal steering, is summarised in Appendix B.

4 Results

4.1 Experimental setup

We evaluate Qwen-2.5-7B (Yang et al., 2024) and Llama-3.1-8B (Grattafiori et al., 2024) on four open-ended QA datasets: TruthfulQA (Lin et al., 2022), TriviaQA (Joshi et al., 2017), SciQ (Welbl et al., 2017), and NQ-Open (Kwiatkowski et al., 2019). For each question we take the greedy answer and label it with BLEURT (Sellam et al., 2020) against the reference, using the thresholds of TSV (Park et al., 2025) (0.5 for TruthfulQA/TriviaQA, 0.2 for SciQ/NQ-Open). To stay directly comparable to prior work, we reuse the generations, labels, and the 75%/25% pool/test split released with TSV. The detector is fit on the pool and applied once to the held-out test set; layer and shrinkage are chosen by five-fold cross-validation on the pool alone (§3.3), so the test set is never seen during selection. We report AUROC with a bootstrap standard error over the test set ($B = 300$).

We compare against the baselines reported by TSV (Park et al., 2025), which span output-uncertainty, sampling, self-evaluation, and internal-state detectors (all listed in Table 1), together with TSV † , the fully-supervised variant of their method that trains the steering vector on ground-truth labels. Like TSV † we use the full pool labels, but our method is closed-form and trains nothing.

4.2 Main results

Table 1 reports AUROC for both models on all four datasets. The closed-form detector matches or exceeds the fully-supervised TSV † on five of eight settings while training nothing, and it never ranks

Method	Qwen-2.5-7B				Llama-3.1-8B			
	TruthfulQA	TriviaQA	SciQ	NQ Open	TruthfulQA	TriviaQA	SciQ	NQ Open
Perplexity	65.1	50.2	53.4	51.2	71.4	76.3	52.6	50.3
LN-Entropy	66.7	51.1	52.4	54.3	62.5	55.8	57.6	52.7
Semantic Entropy	66.1	58.7	65.9	65.3	59.4	68.7	68.2	60.7
Lexical Similarity	49.0	63.1	62.2	61.2	49.1	71.0	61.0	60.9
EigenScore	53.7	61.3	63.2	57.4	45.3	69.1	59.6	56.7
SelfCheckGPT	61.7	62.3	58.6	63.4	57.0	80.2	67.9	60.0
Verbalize	60.0	54.3	51.2	51.2	50.4	51.1	53.4	50.7
Self-evaluation	73.7	50.9	53.8	52.4	67.8	50.9	54.6	52.2
CCS	67.9	53.0	51.9	51.2	66.4	60.1	77.1	62.6
HaloScope	81.3	73.4	76.6	65.7	70.6	76.2	76.1	62.7
SAPLMA†	81.7	82.0	81.5	67.9	78.2	83.7	77.3	62.8
TSV†	88.7	84.2	84.8	76.2	85.5	87.2	88.6	78.0
Ours (gradient-free)	90.1 ± 2.9	91.6 ± 0.5	86.2 ± 2.3	<u>75.3</u> ± 1.7	<u>82.7</u> ± 2.9	90.9 ± 0.6	<u>87.6</u> ± 2.3	78.9 ± 1.4

Table 1: Hallucination-detection AUROC (%). Baseline and TSV† numbers are from [Park et al. \(2025\)](#); † marks fully-supervised methods. Best per column in **bold**; underline marks Ours ranking second. For DTS (Ours), steering is applied on Llama-3.1-8B for TruthfulQA ($\alpha = 16$), SciQ ($\alpha = 4$), and NQ Open ($\alpha = 8$). All remaining DTS results use $\alpha = 0$ (no steering). These strengths are selected by cross-validation.

lower than second: only TSV† ever places above it, and on the three settings where it trails, no other method comes between the two. Two of those gaps are within a single bootstrap standard error (Qwen NQ-Open, -0.9 ; Llama SciQ, -1.0). A paired bootstrap against our own from-scratch TSV† reproduction (Appendix C) puts this on firmer footing: at its unsteered operating point, DTS wins six of eight settings, two of them significantly (Qwen TriviaQA, Qwen NQ-Open), and loses one significantly (Llama TriviaQA). The one real shortfall is Llama TruthfulQA, where the truthful and hallucinated states overlap and no linear readout can split them, the regime in which a steering step recovers the lost separation (§4.4): the 82.7 reported here is that steered value, since an unsteered readout alone reaches only 80.7 (Table 3). Against the other method that uses the full labels, the supervised probe SAPLMA†, the discriminant wins in every column.

4.3 The readout is the bottleneck

The remaining sections trace the gain to its source and chart where the readout runs out. We start by holding the whole pipeline fixed and varying only how a layer’s activation is scored (Table 2): from TSV’s cosine-to-centroid, through the unwhitened difference of class means, to whitening by the within-class covariance, across all eight model×dataset settings. Whitening accounts for essentially the entire improvement over cosine in every setting, with a mean gain of $+7.8$ AUROC (range $+4.1$ to $+10.7$ across settings), while the un-

whitened difference of means stays within a point of cosine in seven of eight settings. The direction is not what matters; the metric is. The full and diagonal whitened forms are comparable throughout, differing by at most 0.6 AUROC, so the gain comes from rescaling each axis by its within-class variance rather than from the off-diagonal covariance; we keep the full discriminant as the general case. The effect is larger on Llama on average (mean gain $+8.5$ vs. $+7.2$ on Qwen), whose activations are more strongly correlated and where an isotropic cosine therefore discards more of the separating signal.

A related question is whether this gain is specific to our closed-form Fisher formula or to covariance-awareness more generally. We swap the discriminant for three standard classifier heads — linear SVM, regularized LDA, and ridge logistic regression — under the same outer protocol (same PCA subspace, same 5-fold CV independently selecting each head’s own layer and hyperparameter; full results in Appendix E). Regularized LDA, another covariance-aware head, ties our discriminant almost exactly (84.93 vs. 84.93 mean AUROC, within 0.2 points in seven of eight settings); linear SVM, which ignores the covariance structure, is the clear underperformer (82.84 mean), worst on Qwen/SciQ (-9.96) and Qwen/TruthfulQA (-5.42); ridge logistic regression sits in between (84.07). The mechanism, not the specific formula, drives the gain: any covariance-aware head recovers it.

Model	Dataset	Ref. (ℓ^* , λ^*)	Cosine	Diff-of-means	Diag. whiten	Full Mahalanobis
Qwen-2.5-7B	TruthfulQA	(15, 0.3)	86.0 $\pm$ 3.7	85.9 $\pm$ 3.8	89.5 $\pm$ 3.2	90.1 $\pm$ 2.9
Qwen-2.5-7B	TriviaQA	(17, 0.1)	81.9 $\pm$ 0.9	81.1 $\pm$ 0.9	91.5 $\pm$ 0.5	91.6 $\pm$ 0.5
Qwen-2.5-7B	SciQ	(17, 0.3)	77.7 $\pm$ 3.0	78.2 $\pm$ 3.0	86.1 $\pm$ 2.3	86.2 $\pm$ 2.3
Qwen-2.5-7B	NQ-Open	(20, 0.1)	68.9 $\pm$ 1.7	68.0 $\pm$ 1.7	75.4 $\pm$ 1.7	75.3 $\pm$ 1.7
Llama-3.1-8B	TruthfulQA	(9, 0.3)	75.2 $\pm$ 3.9	71.7 $\pm$ 4.4	80.6 $\pm$ 3.3	80.7 $\pm$ 3.3
Llama-3.1-8B	TriviaQA	(13, 0.1)	82.6 $\pm$ 0.8	82.7 $\pm$ 0.8	90.2 $\pm$ 0.6	90.6 $\pm$ 0.6
Llama-3.1-8B	SciQ	(12, 0.3)	76.9 $\pm$ 3.1	77.5 $\pm$ 3.0	86.9 $\pm$ 2.5	86.7 $\pm$ 2.5
Llama-3.1-8B	NQ-Open	(13, 0.3)	67.5 $\pm$ 1.6	67.1 $\pm$ 1.6	78.1 $\pm$ 1.4	78.2 $\pm$ 1.5
Mean over all 8 settings			77.08	76.52	84.78	84.93

Table 2: Readout ablation, all 8 model $\times$ dataset settings (test AUROC %, bootstrap SE as subscript). Reference (ℓ^* , λ^*) is the CV-selected operating point behind Table 1, held fixed while the readout varies. Bold marks the best readout per row.

Model	Dataset	None ($\alpha=0$)	CAA mean-diff (Ours)	Random (matched norm)	Trained TSV vector ^R
Qwen-2.5-7B	TruthfulQA	90.1 $\pm$ 2.9	90.1 $\pm$ 2.9 ($\alpha = 0$)	87.8 $\pm$ 3.5	89.0 $\pm$ 3.1
Qwen-2.5-7B	TriviaQA	91.6 $\pm$ 0.5	91.6 $\pm$ 0.5 ($\alpha = 0$)	88.7 $\pm$ 0.6	90.2 $\pm$ 0.6
Qwen-2.5-7B	SciQ	86.2 $\pm$ 2.3	86.2 $\pm$ 2.3 ($\alpha = 0$)	83.8 $\pm$ 2.5	79.8 $\pm$ 2.9
Qwen-2.5-7B	NQ-Open	75.3 $\pm$ 1.7	75.3 $\pm$ 1.7 ($\alpha = 0$)	75.0 $\pm$ 1.6	73.2 $\pm$ 1.6
Llama-3.1-8B	TruthfulQA	80.7 $\pm$ 3.3	82.7 $\pm$ 2.9 ($\alpha = 16$)	81.9 $\pm$ 3.1	81.4 $\pm$ 3.0
Llama-3.1-8B	TriviaQA	90.6 $\pm$ 0.6	90.9 $\pm$ 0.6 ($\alpha = 0$)	90.5 $\pm$ 0.6	91.9 $\pm$ 0.6
Llama-3.1-8B	SciQ	86.7 $\pm$ 2.5	87.6 $\pm$ 2.3 ($\alpha = 4$)	86.8 $\pm$ 2.3	84.1 $\pm$ 2.6
Llama-3.1-8B	NQ-Open	78.2 $\pm$ 1.5	78.9 $\pm$ 1.4 ($\alpha = 8$)	78.9 $\pm$ 1.5	79.0 $\pm$ 1.6
Mean over all 8 settings		84.93	85.42	84.16	83.56

Table 3: Steering-direction ablation, all 8 settings (test AUROC %, bootstrap SE as subscript). None uses $\alpha = 0$; CAA and Random use strengths $\alpha \in \{0, 4, 8, 16\}$ selected by cross-validation (CV) on layers after the steering layer. Trained TSV vector^R results are from our reproduction, with the vector applied at its trained strength (no α to select). Bold marks the best arm per row.

4.4 When steering helps, and why

The readout matches or beats TSV $\dagger$ across Qwen and on Llama TriviaQA and NQ-Open, but trails on Llama TruthfulQA and SciQ, and the cause is geometric. We project held-out activations onto the optimal linear direction (fit on the pool only) and measure two standard separability quantities: the signal-detection sensitivity d' , the gap between class means in standard-deviation units, and the overlap coefficient OVL, the shared area of the two score distributions. On TruthfulQA Qwen’s classes separate cleanly ($d'=1.97$, $OVL=0.35$) while Llama’s sit almost on top of one another ($d'=1.29$, $OVL=0.52$; Figure 2). No linear readout can split overlapping clouds, which is exactly where reshaping the representation should help.

Steering does that, and it need not be trained. We add the difference of class means at an early layer to the residual stream, sweep the strength $\alpha \in \{0, 4, 8, 16\}$, and let cross-validation pick α like any other hyperparameter. Table 3 extends the comparison to all eight settings, including our reproduced results for TSV’s *trained* steering vector,

and examines whether the direction matters beyond the presence of a perturbation. On Qwen, whose representation already separates, cross-validation selects $\alpha=0$ for the closed-form direction in every one of its four datasets — steering never helps — while a random direction of matched norm is clearly worse than not steering throughout, and TSV’s trained vector, which has no option to opt out, is on average the most harmful of the three interventions there. On Llama, steering helps in all four datasets: the closed-form direction beats a random matched-norm control by a clear margin in three of four and is a near-tie on the fourth (NQ-Open), so the gain is specific to the direction rather than to the size of the perturbation. The closed-form approach achieves higher AUROC than TSV’s *trained* steering vector in six of eight settings, with a higher overall mean (85.42 vs. 83.56). However, cross-validation selects $\alpha = 0$ on Qwen-2.5-7B, so those results reflect the unsteered configuration. On Llama-3.1-8B, where nonzero steering is selected, the closed-form direction performs comparably to trained TSV steering, suggesting that useful steer-

ing directions can also be obtained directly from class means.

This asymmetry is not an artefact of a single strength or layer. Figure 3 sweeps the steering strength α on Qwen, the model where steering is harmful: cross-validated AUROC is highest at $\alpha=0$ and degrades as the perturbation grows, and the partial rebound at large α comes only from the readout drifting to a late layer the perturbation leaves intact, and it never reaches the unsteered score. Sweeping the steering layer at fixed α tells the same story, with no layer recovering the unsteered readout, so cross-validation selects $\alpha=0$ throughout. Whether steering helps is therefore a property of the model’s representation geometry, read off d' and OVL, rather than a knob to be tuned.

4.5 Representation choice: last token versus pooling

Every result above scores an answer from its *last* token. We test this choice by also pooling every answer-token position (mean, max) and by a training-free per-token-max variant, each independently refit; the full method is in Appendix F. Table 4 reports all 8 settings for both DTS and TSV $\dagger$: for **DTS** the pattern tracks the same class-overlap geometry that governs steering (§4.4) — pooling helps only where the classes overlap (Llama), and is neutral-to-harmful on Qwen, where the representation already separates, with the largest single gain lifting Llama/TruthfulQA past its steered number (Table 3) with no intervention at all — but *not* for TSV $\dagger$, whose gradient-trained last-token readout wins on average for both models instead. Per-token-max is a clear negative result for both detectors throughout. Appendix F discusses both patterns in full.

4.6 Long-form generation

We test transfer to free-form, multi-claim text on FActScore-style biographies (Min et al., 2023): the full 500-entity FActScore pool per model (499 usable after dropping one biography per model with no parseable atomic facts), generated fresh from the same two base models, with a document-level label from atomic-fact verification against Wikipedia (full setup in Appendix G). Table 5 shows the signal survives this shift: DTS clears well above chance on both models and beats three simple training-free uncertainty baselines by a wide margin. Unlike the short-answer case (§4.5), pooling (mean or max) clearly beats the last token on *both* models here

(88.0/89.4 Qwen, 89.5/89.3 Llama, vs. 74.8/84.4 last-token), gaps well outside either pooling variant’s own bootstrap SE; *which* of mean or max wins is not, since the two are within a point of each other and inside their own SEs on both models.

Perplexity and LN-entropy are the stronger two baselines, but DTS’s best pooling still beats them by 17–18 points, well outside every relevant SE; lexical similarity is barely above chance on Qwen (52.6). With $n \approx 499$ per model and a document-level label, this result is an existence proof that the signal transfers, not a precision estimate on par with the short-answer datasets; Appendix G gives the full discussion and stated limitations.

5 Cost versus Accuracy

A detector is worth running only if it costs little next to the model it monitors. Figure 4 places every detector by wall-clock cost and mean AUROC over the four datasets, on both backbones, and is the object of this section; Table 6 in Appendix A lists the numbers behind it. The baselines fall into three groups: sampling methods that draw $K=10$ generations per input, single-pass methods, and the trained steering detector TSV (Park et al., 2025). We time DTS as it is deployed, one forward pass plus the closed-form readout, and count its one-time cross-validated selection separately, as we do the matching search for every baseline.

DTS sits alone in the upper-left of Figure 4: as cheap as the cheapest baselines and more accurate than all of them. It reaches 85.8 AUROC on Qwen and 85.0 on Llama from a single forward pass in about a minute, the budget of perplexity or a trained probe.

The sampling detectors sit far to the right. Ten generations per input cost them several hours over the four datasets, two to three orders of magnitude more than DTS, and none exceeds 66.3 AUROC. Semantic entropy is the most expensive of all and still trails DTS by more than twenty points.

TSV $\dagger$ is the nearest competitor on accuracy (83.5/84.8) and the comparison that matters most. DTS matches or beats it while costing about a third as much even with the steering sweep, learning no parameters, and running at inference memory (~ 17.5 GB) rather than the training-scale footprint TSV $\dagger$ needs (~ 29.7 GB). The lightweight probes share DTS’s single-pass budget but not its accuracy: DTS lifts mean AUROC well above SAPLMA (78.3/75.5) and CCS (56.0/66.5).

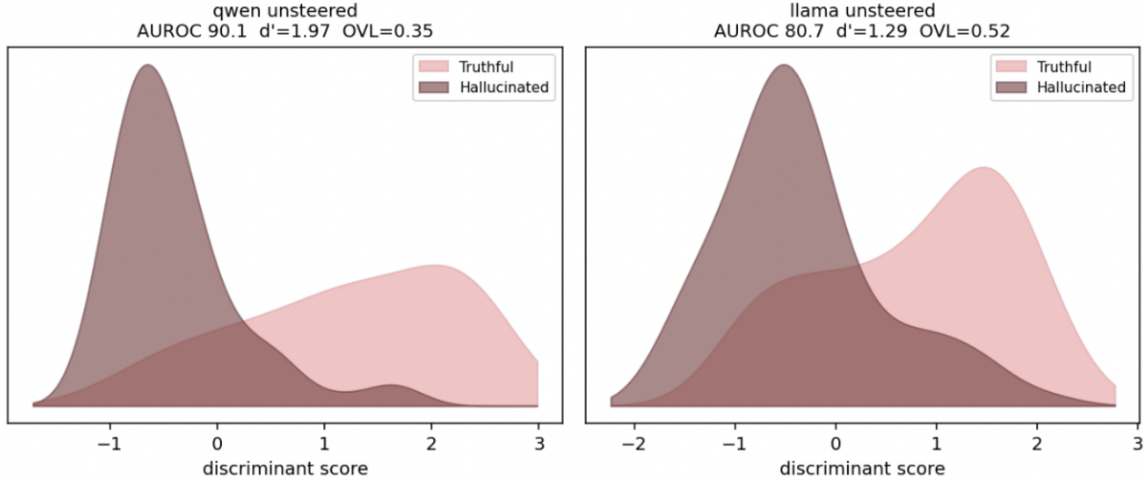


Figure 2: Held-out discriminant-score distributions on TruthfulQA. (a) On Qwen the classes separate ($d'=1.97$, $OVL=0.35$); (b) on Llama they overlap ($d'=1.29$, $OVL=0.52$), so a linear readout alone cannot resolve hallucination.

Model	Dataset	DTS				TSV $\dagger$			
		Last (Ref.)	Mean	Max	Per-token-max	Last	Mean	Max	Per-token-max
Qwen-2.5-7B	TruthfulQA	90.1 ± 2.9	87.7 ± 3.4	88.0 ± 3.2	89.2 ± 3.2	89.6	86.4	84.9	84.7
Qwen-2.5-7B	TriviaQA	91.6 ± 0.5	91.6 ± 0.5	91.4 ± 0.5	84.6 ± 0.8	89.9	88.8	89.5	78.5
Qwen-2.5-7B	SciQ	86.2 ± 2.3	84.5 ± 2.4	86.2 ± 2.2	68.6 ± 3.2	83.0	82.7	81.3	71.9
Qwen-2.5-7B	NQ-Open	75.3 ± 1.7	76.4 ± 1.6	75.5 ± 1.6	66.4 ± 1.8	70.4	71.3	71.6	49.9
Llama-3.1-8B	TruthfulQA	80.7 ± 3.3	85.1 ± 2.8	86.0 ± 2.6	81.7 ± 3.1	80.2	83.6	82.8	79.6
Llama-3.1-8B	TriviaQA	90.6 ± 0.6	93.1 ± 0.5	92.9 ± 0.5	83.3 ± 0.9	92.2	92.1	91.1	84.1
Llama-3.1-8B	SciQ	86.7 ± 2.5	87.9 ± 2.1	87.3 ± 2.2	57.1 ± 3.9	86.1	83.3	78.1	65.0
Llama-3.1-8B	NQ-Open	78.2 ± 1.5	80.6 ± 1.5	79.8 ± 1.5	72.1 ± 1.6	79.2	77.9	78.6	76.7
Mean, all 8 settings		84.92	85.85	85.89	75.38	83.81	83.26	82.23	73.79
Mean, Qwen only		85.79	85.04	85.28	77.20	83.21	82.29	81.81	71.26
Mean, Llama only		84.06	86.67	86.49	73.56	84.41	84.23	82.65	76.32

Table 4: Representation ablation for DTS and TSV $\dagger$, all 8 settings (test AUROC %; DTS carries a bootstrap SE, TSV $\dagger$ does not — see §4.5). Bold marks the best representation per row, per detector.

The gap in Figure 4 is structural, not a matter of tuning. A single forward pass and closed-form linear algebra replace repeated decoding or gradient training, so DTS adds only a small term to inference while the accurate baselines multiply it by a sample or epoch count.

6 Conclusion

We presented DTS, a gradient-free hallucination detector that ranks first in five of eight QA settings and second in the remaining three. Readout ablations identify within-class variance normalization as the main source of improvement over cosine scoring, with a mean gain of 7.8 AUROC points. Comparable performance from diagonal whitening and regularized LDA supports covariance-aware scoring as the key mechanism, rather than an ad-

vantage unique to our formulation.

Steering and token aggregation provide complementary improvements. Cross-validation selects nonzero mean-difference steering in three Llama settings and no steering on Qwen; on Llama, the closed-form approach performs comparably to our reproduced trained TSV steering. Pooling improves DTS across the Llama QA settings, while last-token scoring remains strongest on average for Qwen and reproduced TSV. Per-token maximum scoring generally underperforms, indicating that token-wise application of a last-token readout does not reliably recover useful local evidence.

On long-form biographies, pooled DTS outperforms last-token scoring and the evaluated uncertainty baselines on both models, extending its applicability beyond short answers. Single-pass infer-

Model	n	Halluc. rate	DTS pooling			Training-free baselines		
			Last	Mean	Max	Perplexity	LN-Entropy	Lex. Similarity
Qwen-2.5-7B	499	89.6%	74.8 $\pm$ 4.1	88.0 $\pm$ 3.1	89.4 $\pm$ 2.3	71.0 $\pm$ 4.2	69.3 $\pm$ 4.1	52.6 $\pm$ 4.2
Llama-3.1-8B	499	71.1%	84.4 $\pm$ 2.0	89.5 $\pm$ 1.5	89.3 $\pm$ 1.4	69.0 $\pm$ 3.0	72.5 $\pm$ 2.2	65.9 $\pm$ 2.8
Mean, both models			79.6	88.7	89.4	70.0	70.9	59.3

Table 5: Long-form (FActScore-style biography) hallucination detection, full 500-entity pool, nested 5-fold CV AUROC (%), bootstrap SE as subscript). “Halluc. rate” is the fraction of biographies labeled hallucinated under the majority-vote rule. Bold marks the best score per row.

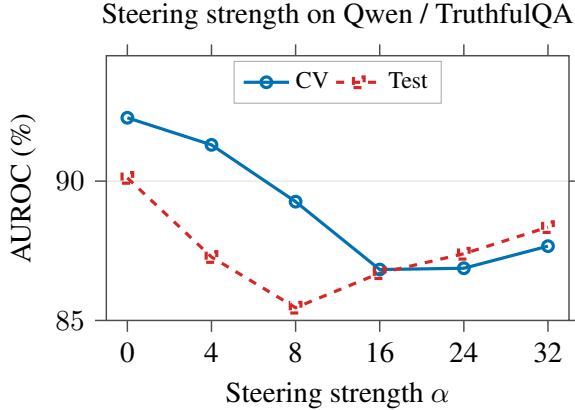


Figure 3: Steering strength on Qwen / TruthfulQA. CV AUROC peaks at $\alpha=0$, selecting no steering. Larger α shifts the readout to later, unperturbed layers, yielding a partial rebound that remains below the unsteered score (90.10).

ence and closed-form fitting also yield lower measured runtime and memory costs than trained TSV. Together, these findings show that effective hallucination detection can be built by carefully scoring and aggregating frozen representations, with selective steering when beneficial, without gradient optimization.

Limitations

Cross-dataset generalisation. DTS fits a discriminant on a labeled pool and applies it once to a held-out test set drawn from the same distribution. When the training and test datasets differ, performance can degrade substantially. Table 10 (Appendix I) reports transfer AUROC for both models across all four dataset pairs. Although several off-diagonal entries are strong (for instance, training on TriviaQA and testing on SciQ yields 81.0 on Llama and 82.8 on Qwen), the mean transfer gap is 14.8 points on Llama and 12.8 on Qwen relative to the in-domain diagonal. The worst cases are pronounced: Llama trained on NQ-Open scores only 55.9 on TruthfulQA, and Qwen trained on Truth-

fulQA scores only 51.7 on SciQ. The selected PCA subspace, optimal layer, and discriminant direction are all adapted to the pool distribution, so a systematic shift in question type, answer length, or factual domain can move the truthfulness signal to a different layer or axis. A practitioner deploying DTS on a new dataset should therefore collect a representative labeled pool from that target distribution; zero-shot transfer across substantially different domains cannot be assumed.

Dependence on the BLEURT labeller. Truthfulness labels follow Equation 2, thresholding BLEURT similarity against a single gold reference at $\tau=0.5$ for TruthfulQA and TriviaQA and $\tau=0.2$ for SciQ and NQ-Open, matching TSV (Park et al., 2025) for direct comparability. This introduces two failure modes. First, BLEURT is a surface-level metric: a correct answer phrased differently from the reference may be labelled hallucinated, while a fluent near-paraphrase of a wrong answer may pass. Second, the threshold is fixed across all questions in a dataset, so answers near the boundary are sensitive to small wording differences rather than factual correctness. Both failure modes introduce label noise that propagates into the discriminant’s class means and covariance estimate, degrading the learned direction. Additional experiments with an LLM-as-a-judge labeller would clarify how much of the remaining error budget stems from label noise versus geometric inseparability of the activations themselves; we leave this for future work.

Steering under limited overlap. The causal steering step (§3.4) operates in a narrow window even in the regime it targets ($d' \lesssim 1.3$, $OV \gtrsim 0.5$). The steering layer and strength $\alpha \in \{0, 4, 8, 16\}$ are chosen by a coarse cross-validated sweep that may miss the optimal intervention, and the decision to steer at all relies on d' and OV estimated from a finite pool, which can be unreliable when labeled data is scarce. Our intervention is also restricted

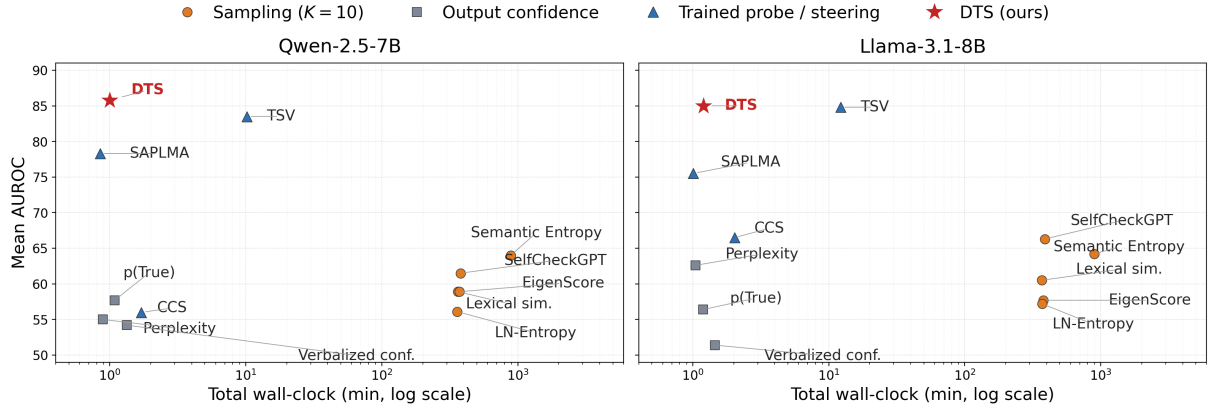


Figure 4: Cost versus accuracy for each detector, on Qwen-2.5-7B (left) and Llama-3.1-8B (right). The x -axis is total wall-clock over the four datasets (log scale); the y -axis is mean AUROC over the same datasets. DTS (red star) is the deployed single-pass detector; its steering-selection sweep is a one-time cost (Table 6). DTS sits alone in the upper-left: accurate and cheap, while the sampling methods (right) and TSV (centre) cost far more for equal or lower accuracy.

to a single residual-stream layer, whereas truthful and hallucinated states may overlap across multiple layers or attention heads; a broader multi-layer search could recover separation that a single-layer intervention misses.

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Appendix

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A Detection cost details

Table 6 reports the per-method detection cost and accuracy summarised in §5 and Figure 4.

B DTS detection algorithm

Algorithm 1 presents the full DTS procedure of §3. All fitted components, including PCA, the whitened discriminant, the calibrator, and any optional steering direction, are estimated using only the training portion of each cross-validation fold. Layer, shrinkage, and optional steering strength are selected using pool-internal validation AUROC. The selected configuration is then refitted on the full pool and evaluated once on the held-out test set. Throughout, $y = 1$ denotes a truthful response, so larger discriminant scores indicate greater truthfulness.

C Significance testing against TSV[†]

This appendix expands §4.2. Table 1 reports point estimates only, and TSV[†]’s published numbers carry no uncertainty measure of their own, so its wins and losses there cannot be read as resolved without a test. The original TSV paper does not retain per-example TSV[†] predictions, so a true paired test needs a fresh TSV[†] run: we train our own from-scratch TSV[†] reproduction (same OT-loss and gated-steering implementation as Park et al. (2025), trained directly on the full pool’s ground-truth labels with checkpoint selection on a pool-internal validation split, never the test set) and recompute DTS’s own per-example test predictions at its unsteered operating point. A paired bootstrap over test indices (1000 resamples, the same

Algorithm 1 DTS hallucination detection

Require: Frozen $\mathcal{M}$; pool $\mathcal{P}$; test set $\mathcal{T}$; folds K
Fit on the pool $\mathcal{P}$

- 1: Extract last-token representations $\mathbf{h}_i^{(\ell)}$ from all layers ▷ Eq. 1
- 2: **for** each candidate (ℓ, p, λ) **do**
- 3: Fit the p -dimensional PCA subspace and regularized discriminant on $\mathcal{P}$ ▷ Eqs. 3–5
- 4: Compute truthfulness scores and fit the probability calibrator ▷ Eq. 6
- 5: Record mean cross-validated AUC
- 6: **end for**
- 7: $(\ell^*, p^*, \lambda^*) \leftarrow \arg \max_{\ell, p, \lambda} \text{AUC}_{\text{CV}}$ ▷ Eq. 7
- Evaluate on the test set $\mathcal{T}$*
- 8: Refit DTS on all of $\mathcal{P}$ using (ℓ^*, p^*, λ^*)
- 9: $\hat{p}_i \leftarrow \Pr(y_i=1 \mid x_i)$ for each $x_i \in \mathcal{T}$
- 10: Report test performance
- Optional causal steering*
- 11: Form truthfulness direction $\mathbf{v}$ and steer at layer ℓ_s ▷ Eq. 8
- 12: Select (ℓ_s, α) on $\mathcal{P}$ and retain the better steered or unsteered detector ▷ §3.4

resampled index set applied to both methods each iteration) then directly estimates the distribution of $\text{AUC}_{\text{DTS}} - \text{AUC}_{\text{TSV}^\dagger}$ under whatever correlation actually holds between the two methods’ errors on shared test items, with no independence assumption. This changes the comparison basis from Table 1 in two ways: DTS is scored unsteered throughout (not at the steered operating point Table 1 uses for Llama TruthfulQA), and TSV[†] is our own reproduction rather than the original paper’s run, so the two tables are not directly comparable row by row.

DTS’s unsteered readout wins six of eight settings against our reproduction; three of the eight results are significant (two wins, Qwen/TriviaQA and Qwen/NQ-Open; one loss, Llama/TriviaQA), and the other five are not distinguishable from noise. Qwen/TriviaQA and Qwen/NQ-Open are where the closed-form readout’s advantage over a trained steering vector is most defensible; Llama/TriviaQA is the one setting where the trained vector wins with confidence, a genuine shortfall alongside the geometric one already discussed in §4.4.

D Selection and robustness

Because we report a single test number, the operating layer and shrinkage must be chosen with-

Method	Grad. train	Time (min) Q/L	FLOPs (PF) Q/L	AUROC Q/L
<i>Multi-generation (sampling), $K=10$</i>				
LN-Entropy	No	358/370	262/277	56.1/57.2
Lexical similarity	No	362/368	262/277	58.9/60.5
EigenScore (Chen et al., 2024)	No	371/379	367/388	58.9/57.7
SelfCheckGPT (Manakul et al., 2023)	No	380/387	469/494	61.5/66.3
Semantic entropy (Kuhn et al., 2023)	No	888/893	316/331	64.0/64.2
<i>Single forward pass</i>				
Perplexity	No	0.9/1.0	11/12	55.0/62.6
Self-eval $p(\text{True})$ (Kadavath et al., 2022)	No	1.1/1.2	13/14	57.7/56.4
Verbalized conf.	No	1.3/1.5	13/13	54.2/51.4
SAPLMA (Azaria and Mitchell, 2023)	Yes	0.8/1.0	10/11	78.3/75.5
CCS (Burns et al., 2023)	Yes	1.7/2.0	21/22	56.0/66.5
TSV (Park et al., 2025)	Yes	10.2/12.3	138/146	83.5/84.8
DTS (single pass)	No	1.0/1.2	10/11	85.8/85.0
DTS (+ α-sweep)	No	3.4/4.1	42/44	85.8/85.0

Table 6: Detection cost and accuracy on Qwen-2.5-7B (Q) / Llama-3.1-8B (L), summed over the four datasets (wall-clock time, PFLOPs, mean AUROC). “Grad. train” marks gradient-trained methods. *DTS (single pass)* is the deployed detector; + α -sweep adds the one-time steering-selection diagnostic (same AUROC).

Model	Dataset	DTS (unsteered)	TSV $\dagger$ (ours)	Diff [95% CI]	Sig.
Qwen-2.5-7B	TruthfulQA	90.12	89.56	+0.55 [−2.31, +3.23]	n.s.
Qwen-2.5-7B	TriviaQA	91.58	89.91	+1.67 [+0.74, +2.57]	sig.
Qwen-2.5-7B	SciQ	86.17	82.98	+3.19 [−0.19, +6.63]	n.s.
Qwen-2.5-7B	NQ-Open	75.32	70.39	+4.93 [+2.31, +7.57]	sig.
Llama-3.1-8B	TruthfulQA	80.74	80.20	+0.54 [−4.07, +4.91]	n.s.
Llama-3.1-8B	TriviaQA	90.58	92.15	−1.57 [−2.42, −0.74]	sig.
Llama-3.1-8B	SciQ	86.73	86.11	+0.63 [−2.57, +3.55]	n.s.
Llama-3.1-8B	NQ-Open	78.17	79.17	−1.00 [−3.13, +1.13]	n.s.

Table 7: Paired bootstrap (1000 resamples), DTS (unsteered) vs. our own from-scratch TSV $\dagger$ reproduction, all 8 settings. CI is the 95% paired-bootstrap interval of the AUROC difference.

Selection	AUROC	Gap to oracle
Fixed paper layer	89.66	1.45
Single validation split	89.73	1.38
Five-fold pool CV (Ours)	90.10	1.01
Oracle layer (upper bound)	91.11	—

Table 8: Selection-protocol ablation on Qwen / TruthfulQA (AUROC %).

out touching the test set. Table 8 compares our five-fold pool cross-validation against a fixed paper layer and a single validation split, with the best-on-test layer as an upper bound. Cross-validation is the strongest honest selector and lands within 1.0 AUROC of the oracle layer, while a single split or a fixed layer give up more.

The detector is otherwise insensitive to its remaining choices, each within a point on Qwen / TruthfulQA. Shrinkage adds about one AUROC over an unregularised covariance and stays within roughly a point across an order of magnitude,

with test AUROC between 89.0 and 90.2 for $\lambda \in [0.1, 1.0]$. Enlarging the discriminant beyond the rank-one Fisher direction does not help either: cross-validated AUROC peaks at $k=1$ (92.28) and is flat or lower for every larger subspace up to $k=16$, so the subspace size is selected as one. For PCA dimension p , cross-validated AUROC peaks at $p=200$ and remains nearly flat thereafter (Figure 5), so CV selects $p=200$ without consulting the test set. The test curve then declines at larger p , confirming that subspaces beyond $p=200$ overfit the pool rather than generalise; that decline is a post-hoc diagnostic, not the basis for the choice.

E Alternate discriminant heads

This appendix expands §4.3. Table 9 replaces the closed-form Fisher discriminant with three standard classifier heads under the identical outer protocol: PCA to the same $p=200$ dimensions, the same 5-fold stratified CV independently selecting each head’s own best (layer, hyperparam-

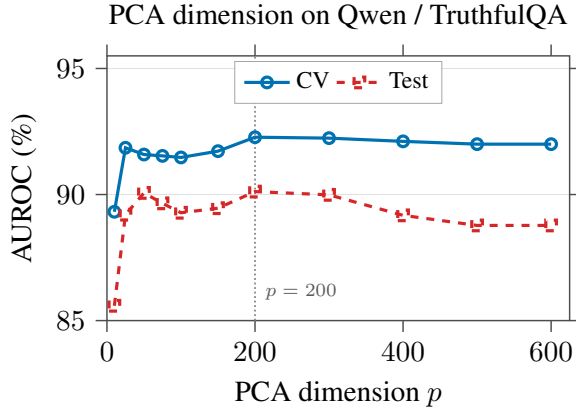


Figure 5: Effect of PCA dimension on Qwen / TruthfulQA. Blue circles show five-fold CV AUROC (with shrinkage selected within CV); red squares show held-out test AUROC. CV peaks at $p=200$ (dotted line) and remains nearly flat thereafter. Test performance then declines, signalling overfitting at larger dimensions.

ter) pair, and the same outer logistic calibration and bootstrap SE. *Linear SVM* (LinearSVC, $C \in \{0.1, 1.0\}$) uses its decision-function score; *regularized LDA* (LinearDiscriminantAnalysis, solver lsqr, shrinkage $\in \{0.1, 0.3\}$) shrinks toward a scaled identity in the same spirit as our own whitening, though not numerically identical; *ridge logistic regression* (ℓ_2 -penalized, $C \in \{0.1, 1.0\}$) uses its raw logit before calibration. Only the classifier producing the 1-D discriminant score changes; everything else is identical to every other comparison in this paper.

Regularized LDA, a second covariance-aware head, matches our closed-form discriminant almost exactly, confirming the paper is not overclaiming novelty of the specific formula: a standard, off-the-shelf shrinkage-LDA implementation recovers essentially the same performance. *Linear SVM*, a max-margin classifier with no covariance-awareness, is the clear underperformer, losing the most ground on Qwen/SciQ (-9.96) and Qwen/TruthfulQA (-5.42). *Ridge logistic regression* sits in between, consistently a touch behind but never by much. The fair summary is that covariance-awareness beats a margin-based or vanilla-regularized classifier, not that our specific closed-form formula is uniquely necessary.

F Representation choice: additional discussion

This appendix expands §4.5. Every main-text result scores an answer from its *last* token’s hidden

state. Last-token representations are also used by TSV (Park et al., 2025), SAPLMA (Azaria and Mitchell, 2023), and INSIDE (Chen et al., 2024), and for the attention-head probes in ITI (Li et al., 2023). INSIDE additionally compares last-token representations with mean pooling (Chen et al., 2024). More recent work highlights that fixed token choices can miss informative positions when hallucinations are localized within longer answers (Niu et al., 2025).

We test the representation choice directly: holding everything else fixed, we keep every answer-token position and reduce it four ways — *last* (our own choice), *mean*-pooled, and *max*-pooled, each independently refit by the same 5-fold layer/shrinkage cross-validation as the headline detector, and *per-token-max*, which does no retraining: the already-fit last-token discriminant is applied to every individual token at test time and the maximum score is kept. This tests token-wise scoring without learning an adaptive token selector, as in HaMI (Niu et al., 2025).

Table 4 (main text) puts both detectors side by side, all 8 settings. For **DTS**, the effect tracks the same class-overlap geometry that governs steering (§4.4): on Qwen, whose classes already separate, last-token wins on average (85.79 vs. 85.04/85.28 for mean/max) and pooling buys nothing, while on Llama, where the classes overlap, mean or max pooling beats last-token by 2.4–2.6 points on average (84.06 vs. 86.67/86.49). The single largest gain is Llama/TruthfulQA, the paper’s one real shortfall (§4.2): max-pooling alone lifts DTS’s AUROC from 80.7 to **85.9**, past the steered number for that setting (82.7, Table 3) with no intervention at all, so representation choice may be the cheaper fix there.

For **TSV**[†], our from-scratch reproduction trained directly on the full pool’s ground-truth labels at the same steering layer and on the identical pool/test split as every other comparison in this paper, the pattern is qualitatively different: last-token wins on average for *both* Qwen (83.2 vs. 82.3/81.8) and Llama (84.4 vs. 84.2/82.7), and wins outright in six of eight individual settings. Gradient training on the last token, unlike DTS’s closed-form readout, does not benefit from pooling even where the classes overlap. Per-token-max, though, is a clear negative result for *both* detectors throughout, collapsing performance on every setting and catastrophically so on three for DTS (57.1 on Llama/SciQ, 68.6 on Qwen/SciQ, 66.4 on

Model	Dataset	Ours	Linear SVM	Reg. LDA	Ridge logistic
Qwen-2.5-7B	TruthfulQA	90.1 $\pm$ 2.9	84.7 $\pm$ 3.7	90.2 $\pm$ 2.9	88.4 $\pm$ 3.4
Qwen-2.5-7B	TriviaQA	91.6 $\pm$ 0.5	91.5 $\pm$ 0.5	91.6 $\pm$ 0.5	91.5 $\pm$ 0.5
Qwen-2.5-7B	SciQ	86.2 $\pm$ 2.3	76.2 $\pm$ 3.1	86.4 $\pm$ 2.3	81.4 $\pm$ 2.8
Qwen-2.5-7B	NQ-Open	75.3 $\pm$ 1.7	74.6 $\pm$ 1.7	75.3 $\pm$ 1.7	75.2 $\pm$ 1.7
Llama-3.1-8B	TruthfulQA	80.7 $\pm$ 3.3	81.3 $\pm$ 3.1	80.5 $\pm$ 3.3	80.7 $\pm$ 3.2
Llama-3.1-8B	TriviaQA	90.6 $\pm$ 0.6	90.7 $\pm$ 0.6	90.6 $\pm$ 0.6	90.8 $\pm$ 0.6
Llama-3.1-8B	SciQ	86.7 $\pm$ 2.5	86.9 $\pm$ 2.4	86.8 $\pm$ 2.5	87.2 $\pm$ 2.4
Llama-3.1-8B	NQ-Open	78.2 $\pm$ 1.5	76.8 $\pm$ 1.6	78.1 $\pm$ 1.5	77.4 $\pm$ 1.5
Mean over all 8 settings		84.93	82.84	84.93	84.07

Table 9: Alternate discriminant heads, all 8 settings (test AUROC %, bootstrap SE as subscript), each independently CV-selecting its own (layer, hyperparameter). Bold marks the best head per row.

Qwen/NQ-Open) and comparably for TSV $\dagger$ (73.8 mean, worst on Llama/SciQ at 65.0): naively taking the max over per-token scores from a discriminant fit only to separate last tokens turns any single noisy token into a false positive, confirming this failure mode belongs to naive max-pooled token scoring itself, for either detector’s own readout mechanism, and not to DTS’s closed-form readout specifically — a different failure mode from a *trained* token-selection method.

G Long-form generation: method and discussion

This appendix expands §4.6. A natural concern is whether the closed-form readout transfers to free-form, multi-claim text, where most of a response is correct and only a small span is hallucinated. We evaluate on FActScore-style biographies (Min et al., 2023): all 500 people in FActScore’s released unlabeled entity pool, stratified evenly across its five popularity tiers (the full pool, not a subsample), generated fresh from the same two base models (Qwen-2.5-7B, Llama-3.1-8B) with greedy decoding (repetition penalty 1.3, no-repeat-3-gram, needed to avoid degenerate sentence-repetition loops that appeared in an earlier 130-entity pilot before this constraint was added; the corrected generation settings were used throughout the full 500-entity run), using the prompt below.

Long-form biography generation prompt

Prompt:

Question: Tell me a bio of {topic}.

Answer:

Each biography is decomposed into atomic facts and each fact verified against retrieved Wikipedia passages by an independent judge model (Qwen2.5-32B-Instruct, used only as an evaluator, never as a

subject model), FActScore’s own labeling scheme. Because even the most familiar popularity tier averages 40–58% unsupported facts on these base, non-chat models, “any unsupported fact” labels nearly every biography positive, so we instead label a biography hallucinated iff more than half its atomic facts are unsupported, a document-level rule fixed during an earlier 130-entity pilot, before looking at any AUROC, and then applied unchanged to the full run. DTS is applied completely unmodified, pooling the generation span three ways (last/mean/max) with (ℓ, λ) chosen by nested 5-fold cross-validation for an honest, leak-free estimate ($n=499$ /model after dropping one biography per model with no parseable atomic facts).

Table 5 (main text) shows the signal survives the shift to free-form, multi-claim text: DTS clears well above chance on both models despite a new benchmark, labeling scheme, and judge model. Unlike the short-answer pooling ablation (§4.5), where pooling only helped Llama, **pooling (mean or max) clearly beats the last token on both models here** (Qwen 88.0/89.4 vs. 74.8; Llama 89.5/89.3 vs. 84.4), gaps of 5–14 points that dwarf either side’s own bootstrap SE: a biography’s hallucinated content is typically one or two wrong claims inside an otherwise-correct paragraph, so the last token has no particular reason to fall inside the hallucinated span, while either pooling aggregates over the whole span instead. An earlier 130-entity pilot of this experiment (no bootstrap SEs computed at that scale) read as max-pooling winning clearly over mean on both models; at the full $n\approx 500$ that specific ordering does not hold up, since mean and max are within a point of each other and inside their own SEs on both models (Table 5). The honest claim is *pooling over the last token*, not *max specifically over mean*; the ear-

lier apparent max-over-mean win was evidently small-sample noise. This still contrasts with §4.5’s per-token-max negative result, but the two are not comparable regardless of which pooling wins here: per-token-max there naively reused a last-token-fitted discriminant at every token, while mean/max-pooling here are first-class pooling choices with their own independently-fit discriminants.

Figure 6 makes this concrete with four held-out biographies from the full 500-entity run (real out-of-fold scores $S(x)$ from each model’s own best-performing pooling in Table 5, max for Qwen and mean for Llama, not illustrative numbers): two the majority-vote rule calls truthful and two it calls hallucinated, one pair per model. Qwen’s Dominique Blake biography invents an entire fictional basketball career — a specific but fabricated draft round, pick number, and trade — and DTS assigns it $S(x)=1.00$; its Coco Chanel biography is essentially accurate and scores $S(x)=0.00$. Llama’s Ahsee Tuala biography is fluent and internally consistent but invents the person’s name, nationality, and family entirely, scoring $S(x)=1.00$; its Nikole Hannah-Jones biography has eight supported atomic facts and scores $S(x)=0.00$. In each case the score tracks the *content* of the biography, not its length or fluency, consistent with Table 5’s aggregate finding that document-level fabrication is detectable from the pooled representation alone.

Against the three simple, training-free uncertainty baselines not evaluated elsewhere in this paper, which compares only against supervised probes (mean per-token perplexity, length-normalized predictive entropy over $K=10$ samples, and lexical similarity, $1-\text{mean pairwise ROUGE-L}$, across the same samples), DTS’s best pooling beats all three by a wide margin on both models: Qwen (89.4 vs. the best baseline, perplexity, at 71.0, a +18.4 gap) and Llama (89.5 vs. the best baseline, LN-entropy, at 72.5, a +17.0 gap), both far outside every relevant bootstrap SE. Lexical similarity is the weakest baseline by a wide margin on both models and barely above chance on Qwen (52.6). An earlier 130-entity pilot of this comparison found a real but much smaller gap (+6.9 Qwen, +1.8 Llama); every baseline’s AUROC *dropped* at the full 500-entity scale while DTS’s held roughly steady, consistent with the pilot’s baseline numbers being optimistically noisy at $n=130$ rather than the full run being unusually favorable to DTS. The larger, more reliable sample makes the training-free-baseline comparison more favorable to DTS,

not less.

With $n \approx 500/\text{model}$ and a document-level label, this result is an existence proof that the closed-form readout transfers to long-form, multi-claim text, not a precision estimate on par with the short-answer datasets: this is the full FActScore unlabeled pool, so no larger sample is available from this data source without generating new entities FActScore did not release. The negative class remains thin in percentage terms after majority-vote labeling (52/499, 10.4%, for Qwen) but its absolute count more than doubled over the 130-entity pilot ($21 \rightarrow 52$), which is what matters for 5-fold CV stability. This is document-level detection, not sentence/span-level localization: it shows DTS’s readout correlates with “this document contains predominantly hallucinated content,” not yet “here is the specific hallucinated sentence”; span-level localization is a natural, heavier extension we leave for future work.

H Benchmark details

We reuse the generation and labelling pipeline released by Park et al. (2025) without modification, so our setup is theirs; we restate it here for completeness. Answers are produced by the frozen generator with the prompt below, taking the most-likely answer (greedy decoding, beam search with five beams for the front-end).

Short-answer QA generation prompt

Prompt:

Answer the question concisely.

Q: {question}

A:

Each answer is labelled truthful when its BLEURT similarity to the reference exceeds a threshold τ , set to 0.5 on TruthfulQA and TriviaQA and 0.2 on SciQ and NQ-Open, following Park et al. (2025). For every (model, dataset) pair we take the first 75% of the released index as the development pool $\mathcal{P}$ and the remaining 25% as the held-out test set $\mathcal{T}$; the indices are the TSV front-end’s cached subsample of each dataset (for instance TriviaQA is subsampled to 9,960 of its 17,944 validation rows), and we derive the split from that index rather than from the raw dataset so that no ungenerated example enters the pool.

TruthfulQA (Lin et al., 2022) comprises 817 questions across 38 categories written to elicit imitative falsehoods, where the common human an-

Qwen-2.5-7B (max-pooled)

Coco Chanel ✓ Truthful (GT) $S(x)=0.00$

Gabrielle “Coco” Bonheur Chanel (August 19, 1883 – January 10, 1971) was a French fashion designer who was the founder of the Chanel brand. She was a major figure in twentieth-century fashion and was the first designer to liberate women from the constraints of corsets and crinolines.

Dominique Blake × Hallucinated (GT) $S(x)=1.00$

Dominique “Doom” Blake (born 1970) is an American former professional basketball player. He played college ball at the University of North Carolina and was drafted by the New York Knicks in June, 18th round with pick #324 overall during his rookie draft year before being traded to Seattle SuperSonics for two second-round picks on July 6 that same year.

Llama-3.1-8B (mean-pooled)

Nikole Hannah-Jones ✓ Truthful (GT) $S(x)=0.00$

She is an American journalist and educator. In 2020, she was awarded the Pulitzer Prize for Commentary in recognition of her work on The New York Times Magazine’s “1619 Project”.

Ahsee Tuala × Hallucinated (GT) $S(x)=1.00$

Ahsanullah Talukder, popularly known as “Ahas”, is an Indian Bengali film actor. He was born in Kolkata on 1 January 1972 to the famous theatre artiste and director Sisir Bhaduri (1928–2006) and his wife Sharmila Devi.

Figure 6: Four held-out long-form examples from the full 500-entity run, with real, out-of-fold scores $S(x)$ (probability of hallucination) from each model’s best-performing pooling in Table 5. GT is the document-level majority-vote label from FActScore-style atomic-fact verification, not DTS’s prediction. Both hallucinated examples read as fluently as the truthful ones — the giveaway is factual content the verifier checked against Wikipedia, not surface style.

swer is wrong; we use the generation configuration.

TriviaQA (Joshi et al., 2017) is a large open-domain trivia set with multiple reference answers; we use the closed-book `rc.nocontext` validation split, so the model answers from parametric memory alone.

SciQ (Welbl et al., 2017) contains crowdsourced multiple-domain science exam questions; we use the validation split and score the free-form answer against the reference.

NQ-Open (Kwiatkowski et al., 2019) is the open-domain form of Natural Questions, built from real Google queries answered without a supporting passage; we use the validation split.

We omit the exemplar-sampling, validation-pool, and optimal-transport pseudo-labelling configuration of Park et al. (2025)’s Appendix B: those settings drive their semi-supervised training stage, and our detector is closed-form with no exemplar set or pseudo-labels.

I Cross-dataset transfer

Table 10 examines DTS beyond in-domain evaluation: the discriminant is fit on one dataset’s development pool and applied to another dataset’s held-out test set. Several source–target pairs show

promising transfer. On Qwen, training on TriviaQA or NQ-Open and testing on SciQ yields 82.8 and 82.9 AUROC, respectively, compared with the in-domain score of 86.2. Training on SciQ and testing on TriviaQA also retains substantial discrimination (85.8 versus 91.6 in-domain). On Llama, TriviaQA-to-SciQ transfer reaches 81.0 AUROC. These results suggest that a discriminant fitted on one dataset can capture truthfulness information that remains useful on another.

Transfer is nevertheless uneven. Llama trained on NQ-Open reaches only 55.9 AUROC on TruthfulQA, compared with 82.7 in-domain. On Qwen, TruthfulQA-to-SciQ transfer reaches only 51.7, even after the polarity correction described below, versus 86.2 in-domain. Thus, the selected layer, PCA subspace, and discriminant direction do not consistently generalise across datasets.

An observation concerning TruthfulQA. Our transfer diagnostic revealed a systematic change in score polarity for pairs involving TruthfulQA. With the score orientation fixed on the source dataset, where higher scores indicate truthfulness, raw transfer AUROC fell below chance in 11 of the 12 ordered pairs involving TruthfulQA across the two models. The exception was Llama trained on NQ-Open and tested on TruthfulQA. None of the 12 pairs between TriviaQA, SciQ, and NQ-Open required a polarity reversal. This suggests that the

association between the source-fitted score and target truthfulness can reverse across datasets, rather than simply becoming weaker.

Let A_{raw} denote AUROC (%) with the source orientation retained. Reversing the score yields $100 - A_{\text{raw}}$, and the polarity-corrected diagnostic is

$$A_{\text{flip}} = \max\{A_{\text{raw}}, 100 - A_{\text{raw}}\}.$$

Table 10 reports in-domain results on the diagonal and transfer results off the diagonal. Entries marked with $\ddagger$ use the reversed orientation; for example, Qwen TriviaQA-to-TruthfulQA reaches 78.5 and Llama TriviaQA-to-TruthfulQA reaches 71.6 after correction. Unmarked off-diagonal entries retain the source orientation. Because choosing the reversal from target AUROC uses target labels, the marked entries are an *oracle-polarity diagnostic*, not strictly zero-shot results. The mean transfer gaps of 12.8 points on Qwen and 14.8 on Llama refer to these displayed results; raw source-oriented transfer can degrade more substantially.

Overall, DTS shows potential for cross-dataset detection, but reliable zero-shot deployment remains challenging. These findings motivate lightweight adaptation using a small, separate labeled target set: selecting score polarity, calibrating deployment thresholds, or adapting the discriminant to the target representation distribution. Such approaches could improve deployment reliability, although their effectiveness remains to be evaluated.

J Baseline and implementation details

Baselines. All baseline numbers in Table 1 are reproduced from Park et al. (2025), evaluated under the same generations, labels, and split. *Perplexity* scores an answer by its mean token negative log-likelihood and *LN-Entropy* by its length-normalised predictive entropy. *Semantic Entropy* (Kuhn et al., 2023) clusters multiple samples by meaning and takes the entropy over clusters; *Lexical Similarity* averages pairwise lexical overlap among samples; *SelfCheckGPT* (Manakul et al., 2023) scores consistency between the answer and additional samples. *Verbalize* (Kadavath et al., 2022) reads the model’s stated confidence and *Self-evaluation* its $P(\text{True})$ judgement of its own answer. *EigenScore* (Chen et al., 2024), *CCS* (Burns et al., 2023), *HaloScope* (Du et al., 2024), and the supervised probe *SAPLMA*[†] (Azaria and Mitchell, 2023) are the internal-state detectors; [†] marks methods, like

Table 10: Cross-dataset AUROC (%). Rows are training datasets, columns are test datasets; diagonal entries (in-domain) are in **bold**. $\ddagger$ marks polarity-corrected entries: the score orientation is reversed, yielding $100 - A_{\text{raw}}$ instead of the below-chance raw AUROC. This target-label-informed correction is an oracle diagnostic, not a strictly zero-shot result. Unmarked off-diagonal entries retain the source score orientation.

Llama-3.1-8B				
Train \ Test	TriviaQA	TruthfulQA	SciQ	NQ-Open
TriviaQA	90.9	71.6 $\ddagger$	81.0	63.7
TruthfulQA	71.6 $\ddagger$	82.7	77.7 $\ddagger$	60.4 $\ddagger$
SciQ	80.5	67.8 $\ddagger$	87.6	67.7
NQ-Open	65.7	55.9	78.6	78.9

Qwen-2.5-7B				
Train \ Test	TriviaQA	TruthfulQA	SciQ	NQ-Open
TriviaQA	91.6	78.5 $\ddagger$	82.8	71.8
TruthfulQA	62.4 $\ddagger$	90.1	51.7 $\ddagger$	59.3 $\ddagger$
SciQ	85.8	71.2 $\ddagger$	86.2	70.4
NQ-Open	82.1	77.6 $\ddagger$	82.9	75.3

SAPLMA and TSV[†], that train on the full ground-truth labels.

Implementation. We evaluate Qwen-2.5-7B (Yang et al., 2024) and Llama-3.1-8B (Grattafiori et al., 2024) in half precision with a forward batch size of 16, caching the last-token hidden state at every layer in a single forward pass. The discriminant uses a top- p PCA subspace with $p=200$, a shrinkage coefficient selected from $\{0.1, 0.3\}$, the rank-one Fisher direction ($k=1$), and a class-balanced logistic calibrator. Layer, shrinkage and top- p PCA subspace are chosen by five-fold stratified cross-validation on the pool; uncertainty is a bootstrap standard error over the test set with $B=300$ resamples. The steering vector is the raw difference of class means at the steering layer ℓ_s , added to the residual stream through a forward hook and scaled by $\alpha \in \{0, 4, 8, 16\}$; the readout is then refit on layers strictly after ℓ_s and the higher cross-validated variant is kept. Nothing in the pipeline is trained by gradient descent beyond the two-parameter logistic calibrator.